4	**(a)**	**(i)**	0.225 s <u>and</u> 0.525 s	A1	[1]
		(ii)	period or T = 0.30 s <u>and</u> $\omega = 2\pi / T$	C1	
			$\omega = 2\pi / 0.30$		
			$\omega = 21\ \mathrm{rad\ s^{-1}}$	A1	[2]
		(iii)	speed $= \omega x_0$ or $\omega(x_0^{\ 2} - x^2)^{1/2}$ <u>and</u> $x = 0$	C1	
			$= 20.9 \times 2.0 \times 10^{-2} = 0.42\ \mathrm{m\ s^{-1}}$	A1	[2]

or

use of tangent method:
correct tangent shown on Fig. 4.2 (C1)
working e.g. $\Delta y / \Delta x$ leading to maximum speed in range (0.38–0.46) $\mathrm{m\,s^{-1}}$ (A1)

(b) sketch: reasonably shaped continuous oval/circle surrounding (0,0) B1
curve passes through (0, 0.42) and (0, –0.42) B1
curve passes through (2.0, 0) and (–2.0, 0) B1 [3]